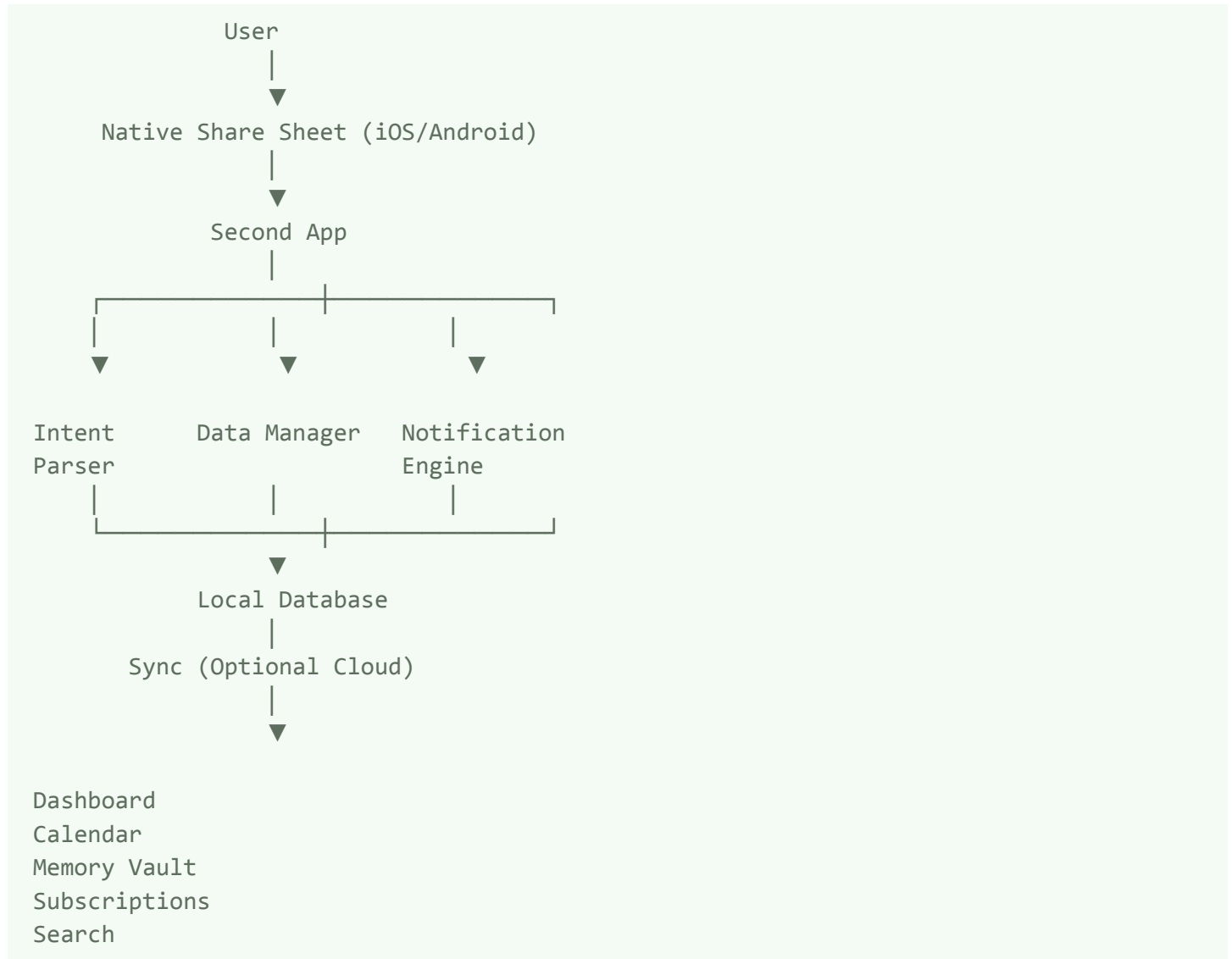


Phase 3 — System Design

1. High-Level Architecture ★★★★★

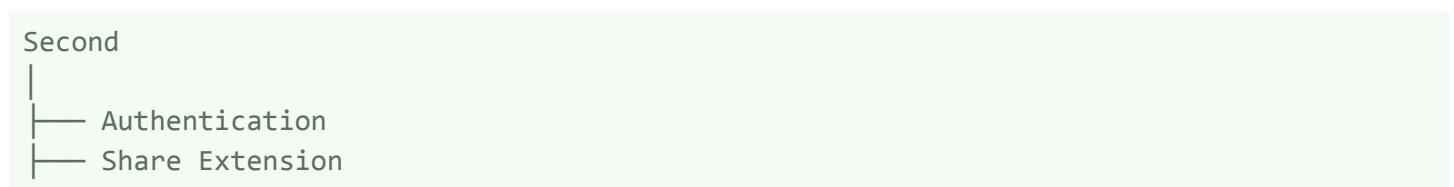
The overall system architecture.



This is the architecture you'll put in your SRS.

2. Component Diagram ★★★★★

Break the app into modules.



- Intent Detection
- Dashboard
- Calendar
- Reminder Manager
- Subscription Manager
- Memory Vault
- Search
- Notification Service
- Settings

Each should be independently buildable.

3. Data Flow Diagram ★★★★★

The most important document.

Example

```
WhatsApp
↓
Share
↓
Second
↓
Extract Information
↓
User chooses action
↓
Store
↓
Dashboard
↓
Notification
Another
Email
↓
Share
↓
Extract Subscription
↓
Store
↓
Subscription Manager
↓
Renewal Reminder
```

This document defines **how information moves**.

4. Information Architecture ★★★★★

How users navigate.

```
Home
|
├── Dashboard
├── Calendar
├── Memories
├── Subscriptions
├── Search
└── Settings
```

5. Database Design ★★★★★

Probably the most important technical document.

```
Tables
-----
Users
Items
Events
Reminders
Subscriptions
Notifications
Tags
```

Each item should have

```
-----
id
title
description
category
source
createdAt
updatedAt
metadata
```

6. API Design ★★★★★

Even if you're not building the backend yet.

Example

```
POST /share
```

```
POST /event
POST /subscription
GET /dashboard
GET /calendar
GET /memories
PATCH /memory
DELETE /memory
```

Thinking through APIs early forces clean boundaries.

7. State Management ★★★★★

Where does data live?

```
Share
↓
Intent
↓
Global Store
↓
Views
Dashboard
Calendar
Memory Vault
Subscriptions
```

8. Sequence Diagrams ★★★★★

These are great in interviews.

Example

```
User
↓
WhatsApp
↓
Share Sheet
↓
Second
↓
Intent Detection
↓
Memory Database
↓
Dashboard Updated
```

Another

```
User
↓
Email
↓
Share
↓
Second
↓
Subscription Parser
↓
Subscription Database
↓
Reminder Engine
```

One thing I would add for Second

Most apps have a **feature-based architecture**.

I think Second should have a **pipeline architecture** because everything starts with one action:

```
Capture
↓
Understand
↓
Structure
↓
Store
↓
Surface
↓
Remind
```

Every feature fits into this pipeline.

For example:

WhatsApp

```
Capture
↓
Meeting
↓
Store Event
↓
Calendar
```

↓
Reminder

Gmail

Capture
↓
Subscription
↓
Store
↓
Renewal
↓
Reminder

Screenshot

Capture
↓
Memory
↓
Store
↓
Search
↓
Retrieve

This pipeline is the core of Second. Instead of designing around features like “Calendar” or “Subscriptions,” you’re designing around the lifecycle of information. Every piece of information follows the same journey—from being captured, to understood, to stored, and finally surfaced when it’s useful. I think this will make both the architecture and the future codebase much cleaner and easier to extend.